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Ekman Spirals with Variable Eddy Viscosity

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1 Introduction

The Ekman spiral is an important consequence of the Coriolis force in meteorology and oceanography. Due to the interaction between shear forces in the medium the wind high up in the atmosphere, or wind over the ocean, will not give rise to a flow in the same direction as at the surface. The adjustment of the flow from the surface spirals and is therefore properly named the Ekman spiral after its discoverer Ekman [1905]. According to Ekman [1905] the surface wind is at an angle from the wind aloft by 45° [Ekman, 1905]. The net flux of the wind is perpendicular to the wind aloft which is known as the Ekman transport and is crucial concept in meteorology and oceanography.

However, a surface wind-turning of 45° assumes a constant viscosity with height, which is unrealistic for a turbulent boundary layer. The boundary layer is around 1–2 km thick and is characterised by a presence of turbulence [Wallace and Hobbs, 2006]. The turbulence primarily arises from solar heating, causing instability and turnover, but also from mechanical turbulence from mean wind decreasing closer to the ground due to frictional shear forces. Although turbulence is non-linear, statistical analysis can provide useful first-order approximations, such as the downward flux being negatively and linearly proportional to the local gradient. Using this proportionality constant one can define the eddy viscosity ν for the boundary layer.

According to the mixing length hypothesis, the eddy viscosity should grow with the distance from the surface as the characteristic scales of the eddies grow. Dimensional analysis of the boundary layer describes a logarithmic relation between the wind speed and height [Wallace and Hobbs, 2006]. This showcases that the eddies, and thus the eddy viscosity, grow with height in the boundary layer. However, after a certain altitude, the turbulent boundary layer ceases to exist, and laminar flow takes over, which rapidly decreases the eddy viscosity. Thus, assuming initial increase of the eddy viscosity followed by decrees at higher altitude is reasonable. Following this logic, the eddy viscosity should in reality vary with altitude, and thus changing the surface wind-turning of 45°.

Analytical models have investigated the effect of depth-dependent eddy viscosities for the Ekman spiral. Grisogono [1995] used a Wentzel–Kramers–Brillouin (WKB) approximation, like in quantum mechanics, for the classical Ekman solution, showcasing that smoothly varying viscosity profiles allow for approximate analytical solutions for the entire Ekman height. Bressan and Constantin [2019] used a direct perturbative method on the viscosity in order to obtain a first-order equation for predicting the surface angle of a constant perturbed viscosity. This was later refined by Constantin [2021] to show that the surface angle is a product of the classical angle and a weighted vertical integral of the perturbation. Some exact analytical solutions have also been derived, including a two-layer model by Dritschel et al. [2020], Guan et al. [2023], an exponential decaying profile by Miles [1994] and monotonically increasing polynomial profiles by Ionescu-Kruse [2021], Guan et al. [2023].

Further insight into the effect of the variable eddy viscosity can be obtained by comparing analytical, numerical solutions and observational data. In this article, a varying eddy viscosity with altitude will be studied under the Ekman equation. This will be done analytically and numerically, and will be compared to reanalysed data. This will give a deeper understanding of the Ekman spiral and provide insight when a surface angle different from 45° can be observed.

2 Theory

2.1 Problem formulation

The Ekman equation is derived from the steady-state Navier–Stokes equations on a rotating frame with shear flow. Assuming laplace friction, incompressible flow, negligible vertical motion, gives the horizontal

momentum equations as

$$\begin{cases} -fv = -\frac{1}{\rho}\frac{\mathrm{d}p}{\mathrm{d}x} + \frac{\mathrm{d}}{\mathrm{d}z}\left(\nu\frac{\mathrm{d}u}{\mathrm{d}z}\right), & \text{(1a)} \\ fu = -\frac{1}{\rho}\frac{\mathrm{d}p}{\mathrm{d}y} + \frac{\mathrm{d}}{\mathrm{d}z}\left(\nu\frac{\mathrm{d}v}{\mathrm{d}z}\right), & \text{(1b)} \end{cases}$$

where u and v are the horizontal velocity components, f is the Coriolis parameter, ρ is the fluid density, and ν is the viscosity. The flow approaches geostrophic balance $(u, v) \to (u_g, v_g)$ far away from the surface, where the friction term in Equation 1 becomes negligible. This problem can be formulated with one equation by introducing a complex velocity as

$$U \equiv u + iv, \tag{2}$$

and a corresponding complex geostrophic velocity. Using Equation 1 and Equation 2 the coupled equations can be combined into the Ekman equation by subtracting the geostrophic balance as

$$\frac{\mathrm{d}}{\mathrm{d}z}\left(\nu\frac{\mathrm{d}U}{\mathrm{d}z}\right) = if(U - U_g), \tag{3}$$

with the boundary conditions

$$U(0) = 0, \tag{4a}$$
$$U(z \to \infty) = U_g. \tag{4b}$$

The surface wind angle can be determined by the argument of the infinitesimal change of the wind at the ground,

$$\theta = \arg\left(\left.\frac{\mathrm{d}U}{\mathrm{d}z}\right|_{z=0}\right). \tag{5}$$

The Ekman transport is defined as the mass flux per unit width for the entire Ekman spiral,

$$T = \int_0^\infty [U - U_g]\,\mathrm{d}z, \tag{6}$$

since the transport represents the net movement of the entire fluid column offset from the geostrophic wind aloft. The angle of the Ekman transport can consequently be determined by the argument, $\theta_T = \arg(T)$.

2.2 The vertical momentum flux

The Ekman equation can be reformulated by introducing a complex vertical momentum flux as

$$\tau = \nu\frac{\mathrm{d}U}{\mathrm{d}z}. \tag{7}$$

Using this definition, differentiating Equation 3 gives

$$\boxed{\frac{\mathrm{d}^2\tau}{\mathrm{d}z^2} = i\kappa^2\tau,} \tag{8}$$

where

$$\kappa^2 \equiv \frac{f}{\nu}. \tag{9}$$

Here, an inverse decay scale κ has been introduced to the system. Equation 8 is similar to equations such as the Helmholtz equation and the time-independent Schrödinger equation, where κ becomes analogous to the potential. The new equation also has a clear physical interpretation, as it describes the vertical structure of the momentum flux in the Ekman problem. Using Equation 4 and 7, the boundary conditions become

$$\left.\frac{\mathrm{d}\tau}{\mathrm{d}z}\right|_{z=0} = -ifU_g, \tag{10a}$$
$$\tau(z \to \infty) = 0. \tag{10b}$$

Since the viscosity is a real positive value the surface angle can be expressed with τ as

$$\theta = \arg\left[\tau(z=0)\right]. \tag{11}$$

The momentum flux directly reduces the integral of the Ekman transport. By inserting Equation 3 into 7 it follows that $\frac{\mathrm{d}\tau}{\mathrm{d}z} = if\left(U - U_g\right)$. Using the boundary condition that the momentum flux vanishes at infinity the Ekman transport reduces to

$$T = \int_0^\infty \left[U - U_g\right] \mathrm{d}z = \frac{i}{f}\tau(z=0), \tag{12}$$

which is proportional to the vertical momentum flux at the surface. Thus, the surface stress at the ground is proportional to the Ekman transport of the fluid column.

2.3 The classical solution

The classical solution to the Ekman equation is obtained when the viscosity is constant with altitude and was found by Ekman [1905]. In this case the solution is

$$\tau(z) = \nu U_g \frac{1+i}{h_{\mathrm{Ek}}} e^{-\frac{1+i}{h_{\mathrm{Ek}}}z}, \tag{13}$$

where

$$h_{\mathrm{Ek}} \equiv \sqrt{\frac{2\nu}{f}} \tag{14}$$

is the Ekman height scale. The surface angle of the classical Ekman theory gives 45°, since the solution is equally real and imaginary at $z = 0$. The transport yields an angle of 90° from the surface wind-turing, meaning that the transport and surface wind is perpendicular.

2.4 The two layer model

The next natural step from a constant viscosity profile is to assume two layers of constant viscosity in each layer. The lower level is labelled as having the viscosity ν_1 up until a height of H, where the next viscosity of ν_2 takes over. From this assumption a superimposed solution similar to Equation 13 is obtained as

$$\begin{cases} \tau_1(z) = A_1 e^{\frac{1+i}{h_{\mathrm{Ek1}}}z} + B_1 e^{-\frac{1+i}{h_{\mathrm{Ek1}}}z}, & \text{(15a)} \\ \tau_2(z) = B_2 e^{-\frac{1+i}{h_{\mathrm{Ek2}}}z} & \text{(15b)} \end{cases}$$

where the Ekman height scales are defined for each layer like in Equation 14. The jump conditions are

$$U_1 = U_2 \quad \text{at } z = H, \tag{16a}$$
$$\tau_1 = \tau_2 \quad \text{at } z = H. \tag{16b}$$

Using the jump conditions together with the boundary conditions in Equation 4 the prefactors become

$$A_1 = -\nu_1 U_g \frac{1+i}{h_{\mathrm{Ek1}}} \cdot \frac{1-\gamma}{1-\gamma+e^{2(1+i)\varphi}\left(1+\gamma\right)}, \tag{17a}$$

$$B_1 = \nu_1 U_g \frac{1+i}{h_{\mathrm{Ek1}}} \cdot \frac{\left(1+\gamma\right)e^{2(1+i)\varphi}}{1-\gamma+e^{2(1+i)\varphi}\left(1+\gamma\right)}, \tag{17b}$$

$$B_2 = \nu_2 U_g \frac{1+i}{h_{\mathrm{Ek2}}} \cdot \frac{2e^{(1+i)\varphi\left(1+\gamma^{-1}\right)}}{1-\gamma+e^{2(1+i)\varphi}\left(1+\gamma\right)}, \tag{17c}$$

$$\text{where } \gamma = \sqrt{\tfrac{\nu_2}{\nu_1}} \tag{17d}$$

and where φ is the non-dimensional layer thickness defined as

$$\varphi = \frac{H}{h_{\mathrm{Ek1}}}. \tag{18}$$

This provides an solution for the vertical momentum flux in the two-layer model, as shown in Figure 1. The figure illustrates how the vertical momentum flux varies with altitude for different viscosity ratios and non-dimensional heights, φ. For an increasing viscosity with height, a larger surface stress in the zonal direction compared to the meridional direction is observed. Conversely, for a smaller upper-layer viscosity, a larger meridional surface stress compared to the zonal surface stress is evident. Observing the interface, H, reveals that for larger φ, the model behaves more like a single-layer model, as the second layer begins at a height where the Ekman spiral has already decayed. Interestingly, smaller values of φ yield a larger difference between the zonal and meridional wind stress.

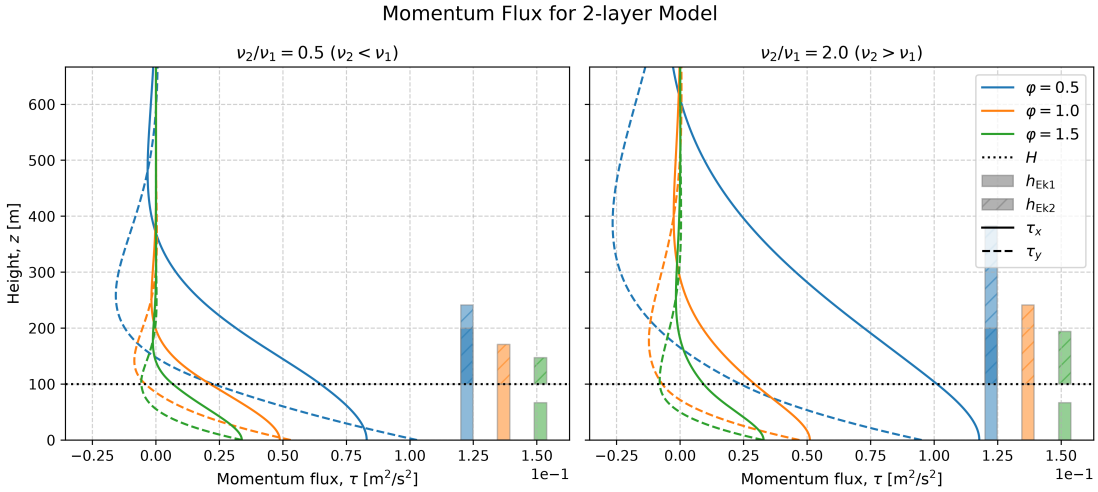


Figure 1: The vertical structure of the momentum flux for the two layer model. The momentum flux depends on the dimensionless parameter $\varphi = H/h_{\mathrm{Ek1}}$ and the ratio of the viscosities. The corresponding Ekman heights and interface are shown with the vertical momentum flux. The solution assumed a purely zonal geostrophic flow $U_g = 10\,\mathrm{m\,s^{-1}}$, a Coriolis parameter of $f = 1 \times 10^{-4}\,\mathrm{s^{-1}}$ and an intersection height of $H = 100\,\mathrm{m}$. The lower viscosities, derived from H and φ, are $\nu_1 = 2\,\mathrm{m^2/s}$, $\nu_1 = 5 \times 10^{-1}\,\mathrm{m^2/s}$, and $\nu_1 = 2.22 \times 10^{-1}\,\mathrm{m^2/s}$ in both figures. In the left figure, ν_2 used half the value of the lower viscosities, and in the right figure, it used double of that value.

By defining the constant

$$R = \frac{1-\gamma}{1+\gamma}. \tag{19}$$

and using the derived solution for the vertical momentum flux, the surface wind angle becomes

$$\theta = \arg\left[(A_1 + B_1)\right] \tag{20}$$

$$= 45^\circ + \arg\left[\frac{e^{(1+i)\varphi} - Re^{-(1+i)\varphi}}{e^{(1+i)\varphi} + Re^{-(1+i)\varphi}}\right], \tag{21}$$

which can be observed as a function of φ in Figure 2. The solutions for different viscosity ratios are shown, half of which have a higher upper-level viscosity and the other half have a lower upper-level viscosity. The solution convergences towards the classical Ekman value of 45° when $\varphi \gg 1$ for all the viscosity ratios. This can be attributed to the high altitude interface H being above the spiral, which reduces the problem to a one layer model.

The zero limit is, however, dependent on the viscosity ratio. If the ratio is finite and fixed, the surface angle approaches 45° as $H \to 0$, again reducing the problem to a one-layer model. But if φ is very small while the viscosity ratio varies, the angle approaches 0° if $\nu_2/\nu_1 \to \infty$, or an angle of 90° if $\nu_2/\nu_1 \to 0$. The surface is thus sensitive to the viscosity ratio when φ is small, as the surface angle can be either above or below 45° depending on the viscosity ratio. When $\varphi \ll 1$, the solution behaves approximately as a linear function between 0 and H, allowing the jump conditions to determine the surface angle. Equation 15 gives

$$\left.\frac{\mathrm{d}\tau}{\mathrm{d}z}\right|_{z=H} \approx -\frac{1+i}{h_{\mathrm{Ek2}}}\tau(H) \implies \frac{H\,|\tau'(H)|}{|\tau(H)|} \sim \frac{\varphi}{\gamma}, \tag{22}$$

which puts a constraint on $\tau(H)$ as h_{Ek2} becomes very small or very large. If the viscosity ratio is very small, $\gamma \ll \varphi$, the momentum flux at the interface is very small, due to the small upper Ekman height

$h_{\mathrm{Ek2}} \ll H$ balancing a finite change in momentum flux. Taylor expanding the momentum flux at the interface gives

$$\tau(H) = \tau(0) + H\tau'(0) + \mathcal{O}\left(H^2\right). \tag{23}$$

Using Equation 10a and neglecting the small terms gives

$$\tau(0) \approx iU_g H \implies \arg[\tau(0)] \approx 90°. \tag{24}$$

Similarly when $\varphi \ll 1$ and $\gamma \gg \varphi$, the change in the momentum flux has to become small, due to large upper Ekman height, $h_{\mathrm{Ek2}} \gg H$ so that the change becomes small to balance the finite momentum flux. Integrating Equation 8 gives

$$\tau'(H) - \tau'(0) = i\frac{fH}{\nu_1}\tau(0) + \mathcal{O}\left(H^2\right) \implies \tau(0) \approx \frac{\nu_1 U_g}{fH} \implies \arg[\tau(0)] \approx 0°. \tag{25}$$

In the case of weaker upper viscosity the perpendicular angle can be interpreted as the entire rotation occurring at the surface since the viscosity becomes negligible aloft. In the opposite case, a weaker lower viscosity allowing minimal spiralling, since the absence stress cannot force a rotation.

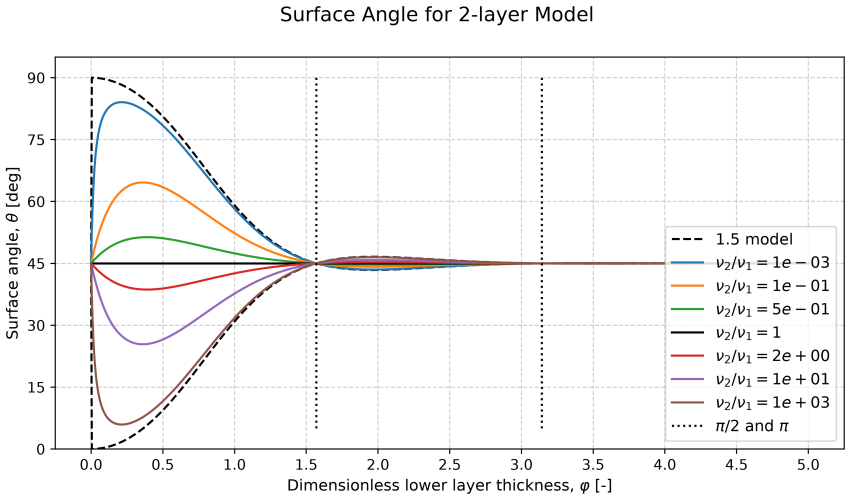


Figure 2: The surface angle as a function of the dimensionless layer thickness $\varphi = H/h_{\mathrm{Ek1}}$ for the two-layer model. Different coloured curves correspond to different layer viscosity ratios. The classical Ekman angle of 45° is shown as a reference.

The full vertical spiral of the two-layer model can be obtained by calculating the wind direction at each height. This spiralling compared to the classical Ekman theory can be observed in Figure 3. If φ is large, then most of the spiral is in the lower layer, and therefore close to the classical Ekman theory. In the other cases it is seen that the angle overshoots in the layer with high viscosity and undershoots in layer with low viscosity compared to the classical solution.

Varying φ and observing the transport can be seen in Figure 4. Although the values deviate from the classical ones when φ becomes small, the ratio between them stays the same compared to the angles in Figure 2, that is 90°. When $\varphi \gg 1$ the solution approaches the classical transport values. However, when $\varphi \to 0$ the transport magnitude depends on whatever the upper level viscosity is. If $\nu_2 \to 0$ then the transport becomes negligible, but if $\nu_2 \to \infty$ then the transport becomes huge. This makes physical sense since $\varphi \ll 1$ means that H is small compared to h_{Ek2}, leading to a upper level viscosity dominance.

2.5 The exponential decaying viscosity model

Another viscosity profile that can provide an analytical solution is an exponentially decaying viscosity with altitude. Here the viscosity is set to decay from an initial viscosity of ν_0, with a decay rate of k as $\nu(z) = \nu_0 \exp(-kz)$. Solving Equation 8 with this viscosity profile can be done by introducing the new variable

$$t = 2(1+i)\,\varphi_{\mathrm{exp}}\, e^{\frac{kz}{2}}. \tag{26}$$

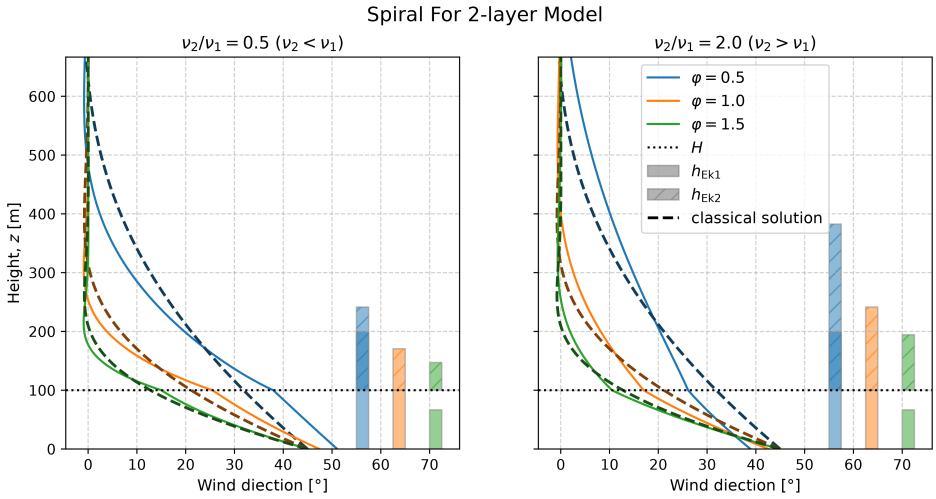


Figure 3: The vertical spiral of the velocities for the two layer model together with the corresponding Ekman heights. The wind direction depends on the dimensionless parameter $\varphi = H/h_{\mathrm{Ek1}}$. The classical Ekman solution (when $\nu_{\mathrm{const}} = \nu_1$) is shown as a reference. The parameters used the same values as in Figure 1.

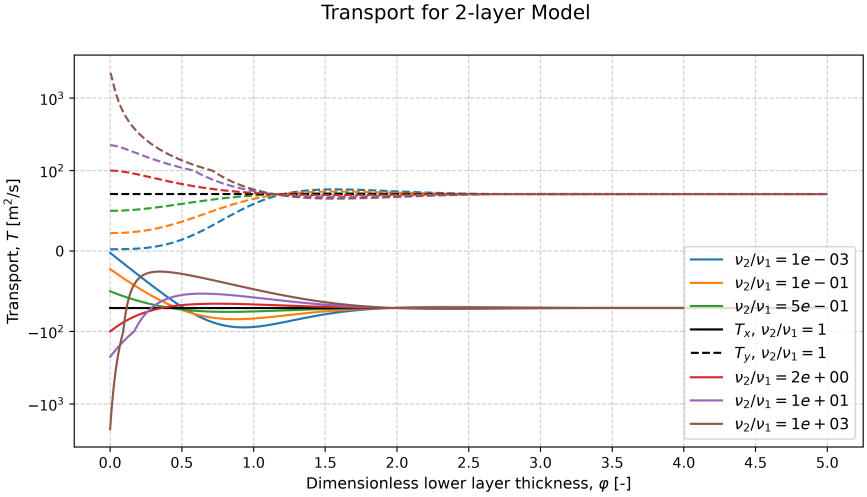


Figure 4: The total transport as a function of the dimensionless layer thickness $\varphi = H/h_{\mathrm{Ek1}}$ for the two-layer model. Different coloured curves correspond to different layer viscosity ratios. The solution assumed $\nu_1 = 0.01\,\mathrm{m^2\,s^{-1}}$.

The non-dimensional parameter φ_{exp} resembles the non-dimensional layer thickness from the two-layer model, but defined as

$$\varphi_{\mathrm{exp}} = \frac{1}{k\,h_{\mathrm{Ek0}}} = \frac{1}{k}\sqrt{\frac{f}{2\nu_0}}. \tag{27}$$

Using these newly defined variables Equation 8 transforms into

$$t^2\frac{\mathrm{d}^2\tau}{\mathrm{d}t^2} + t\frac{\mathrm{d}\tau}{\mathrm{d}t} - t^2\tau = 0, \tag{28}$$

which is a differential equation with a standard solution. The standard solution is a modified Bessel equation as

$$\tau(t) = C_1 I_0(t) + C_2 K_0(t), \tag{29}$$

where C_1 and C_2 are constants, while $I_0(t)$ and $K_0(t)$ are modified Bessel functions of the first and second kind of order zero, respectively. By applying the previously defined boundary conditions from Equation 10 the two constants can be found. Applying the geostrophic boundary condition from Equation 10b shows that $C_1 = 0$ since $I_0(t)$ does not go to zero when $t \to \infty$ (which occurs when $z \to \infty$). Applying the surface boundary condition from Equation 10a an expression for the second constant can be found.

Transforming the solution to depend on the variable z gives

$$\tau(z) = fU_g h_{\mathrm{Ek0}} \frac{1+i}{2} \frac{K_0\left(2\left[1+i\right]\varphi_{\mathrm{exp}} e^{kz/2}\right)}{K_1\left(2\left[1+i\right]\varphi_{\mathrm{exp}}\right)}. \tag{30}$$

This solution for the vertical momentum flux in the exponential model is shown in Figure 5 for different non-dimensional heights, φ_{exp}. For smaller φ_{exp}, the solution is more localized toward the surface, whereas larger φ_{exp} produces a more extended spiral. This implies that for $\varphi_{\mathrm{exp}} \gg 1$, the viscosity changes very little, causing the spiral to behave as in the one-layer model. In contrast, $\varphi_{\mathrm{exp}} \ll 1$ indicate a rapid viscosity decrease from the ground.

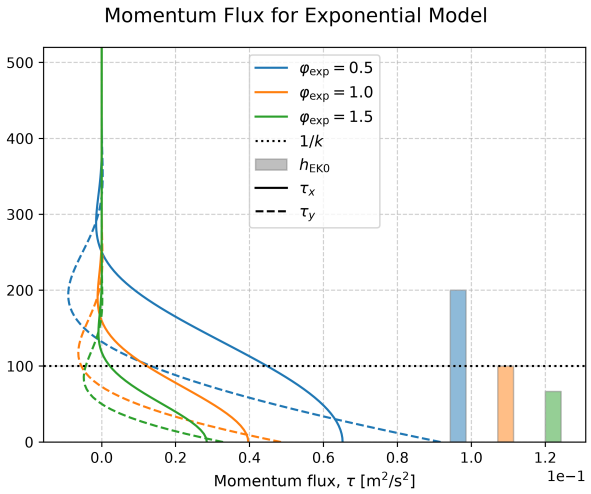


Figure 5: The vertical structure of the momentum flux for exponentially decreasing viscosity together with the corresponding Ekman height and inverse decay rate. The momentum flux depends on the dimensionless parameter $\varphi_{\mathrm{exp}} = 1/k\, h_{\mathrm{Ek0}}$. The solution assumed a purely zonal geostrophic flow $U_g = 10\,\mathrm{m\,s^{-1}}$, a Coriolis parameter of $f = 1 \times 10^{-4}\,\mathrm{s^{-1}}$, a decay rate of $k = 1 \times 10^{-2}\,\mathrm{m^{-1}}$ and initial viscosities of $\nu_0 = 12.5\,\mathrm{m^{-1}}$, $\nu_0 = 0.5\,\mathrm{m^{-1}}$ and $\nu_0 = 0.031\,25\,\mathrm{m^{-1}}$.

Using Equation 30 the surface angle is

$$\theta = 45^\circ + \arg\left[\frac{K_0\left(4\left[1+i\right]\,\varphi_{\mathrm{exp}}\right)}{K_1\left(4\left[1+i\right]\,\varphi_{\mathrm{exp}}\right)}\right], \tag{31}$$

where K_1 is the modified Bessel function of the second kind of order one. Equation 31 is plotted in Figure 6. As this model only provides scenarios where the viscosity decreases from the ground, all angels are greater then the reference angle of 45° (as predicted by the two layer model). One can observe a similar limit as in Figure 2 when $\varphi_{\mathrm{exp}} \to 0$ for $\nu_2 = 0$, as Equation 31 also approaches 90°. The same limit is also true for $\varphi_{\mathrm{exp}} \to 1$, which gives $\theta \to 45°$, although this exponential solution converges slightly slower.

The opposite case when $\varphi_{\mathrm{exp}} \ll 1$ allows for Taylor expansion of $\tau(z)$ about $z = 0$ as

$$\tau(z) \approx \tau_0 + \tau'(0)z + \tau''(0)\frac{z^2}{2} \tag{32a}$$

$$\approx \tau_0 - \underbrace{iU_g z}_{\text{linear}} + \underbrace{\frac{\tau_0 i}{h_{\mathrm{Ek0}}^2} z^2}_{\text{curvature}}. \tag{32b}$$

$$\tag{32c}$$

Comparing the curvature term with τ_0 gives a linear region of $\sim h_{\mathrm{Ek0}}$ when $\varphi \ll 1$. However, at $z \sim h_{\mathrm{Ek0}}$, the momentum flux is approximately zero $\tau \approx 0$ according to Ekman theory. Thus the surface momentum flux has to be imaginary to the leading order to fulfil the vanishing upper boundary condition, providing a perpendicular surface angle of 90° when $\varphi_{\mathrm{exp}} \to 0$.

The wind directions of the exponential model is shown in Figure 5, compared to the classical Ekman theory. The angles are higher than the classical solution at the surface but decreases more rapidly with height, following the logic presented in the two-layer model.

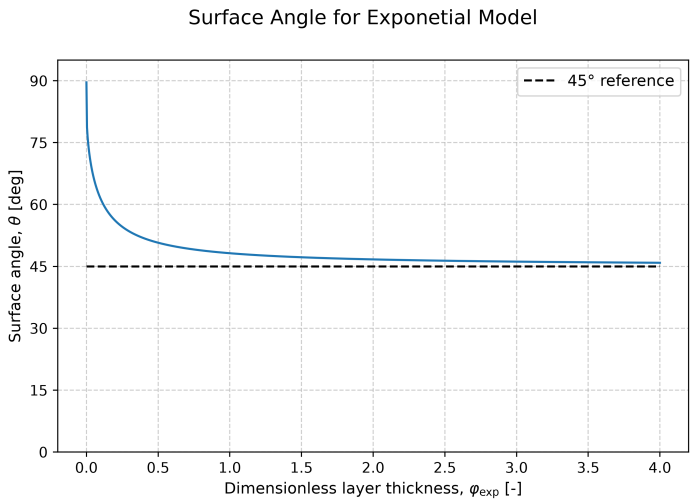


Figure 6: Surface angle for the exponentially decaying viscosity model as a function of the dimensionless parameter $\varphi_{\text{exp}} = H/h_{\text{Ek0}}$. The same parameters as in Figure 5 was used.

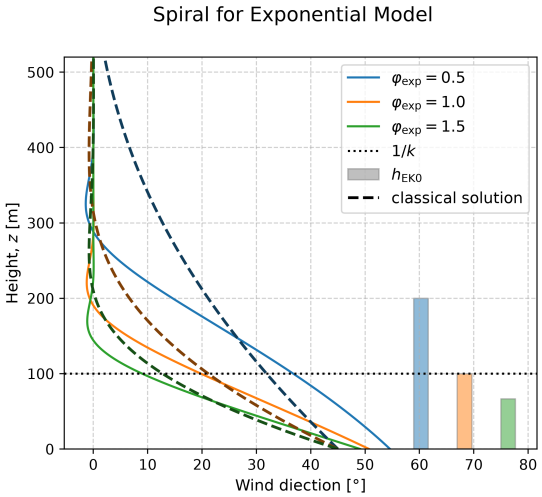


Figure 7: The vertical spiral for the exponentially decreasing viscosity model together with the Ekman height and the inverse decay rate. The wind direction depends on the dimensionless parameter $\varphi_{\text{exp}} = 1/k\, h_{\text{Ek0}}$. The classical Ekman solution (for ν_0) is shown as a reference. The parameters had the same values as in Figure 5.

Varying φ and observing the transport can be seen in Figure 4, presenting similar result to the 2-layer model. When $\varphi \gg 1$, then the transport converges towards finite values, while when $\varphi \to 0$ then transport becomes huge. Again the angle for the transport subtracted by the angles in Figure 6 gives 90°.

2.6 The simplified upper boundary condition

There is difficulty solving Equation 8 due to the altitude dependence of the viscosity $\nu(z)$. Analytical solutions therefore exist only for specific viscosity profiles that allow the equation to be transformed into a solvable form. Equation 8 is furthermore also difficult to solve numerically, as the upper boundary condition is asymptotic. This will put limits on the viscosity profiles to study, as the viscosity cannot increase with altitude continuously. In the physical world, this makes sense, since we assume that high up in the atmosphere (or deep down in the ocean), the flow becomes laminar and the turbulent viscosity small.

A solution to solve this problem is to introduce a simplified upper boundary condition. For the two-layer model, this is reduced to a 1.5 layer model, where we assume that the upper-level viscosity is negligible

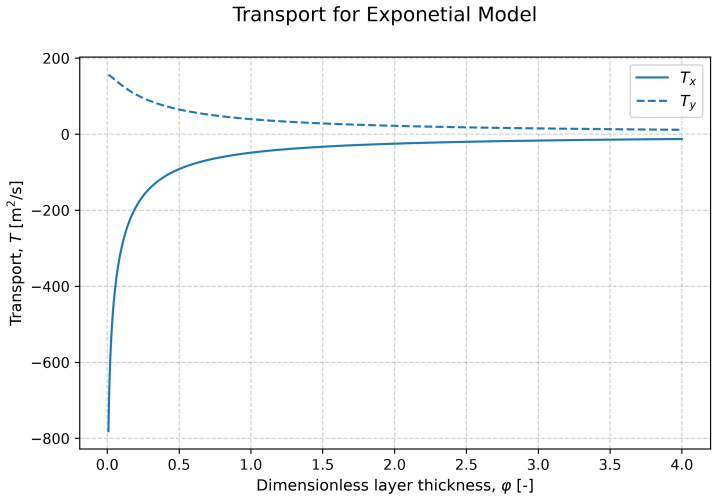


Figure 8: The total transport as a function of the dimensionless layer thickness $\varphi_{\text{exp}} = 1/k\, h_{\text{Ek0}}$ for the exponential model. The solution assumed $k = 0.1\,\text{m}^{-1}$.

($\nu_2 \approx 0$), which corresponds to the upper dashed curve in Figure 2. This also makes physical sense, since we know that both the atmosphere and the ocean consist of a boundary layer where turbulent mixing takes place. Allowing the viscosity to vary freely in the lower layer reduces the problem to a simplified upper boundary condition model. This will allow viscosity profiles that do not converge at $z \to \infty$, since the new domain is between $z \in [0, H_{\text{BL}}]$. This modifies the boundary conditions in Equation 10 as

$$\left.\frac{\mathrm{d}\tau}{\mathrm{d}z}\right|_{z=0} = -ifU_g, \tag{33a}$$

$$\tau(z = H) = 0. \tag{33b}$$

Here, H_{BL} is the height of the boundary layer, and the upper boundary condition was argued from the jump condition in the two-layer model, as seen in Equation 16b.

2.6.1 Some simple viscosity profiles

Using the the simplified upper boundary condition, SUBC model, simple viscosity profiles can be studied and evaluated numerically. The easiest, except for the constant viscosity, is a linearly varying viscosity with height as

$$\nu_{\uparrow}(z) = \frac{\nu_{\max}}{H_{\text{BL}}} z, \tag{34a}$$

$$\nu_{\downarrow}(z) = \frac{\nu_{\max}}{H_{\text{BL}}} \left(H_{\text{BL}} - z\right), \tag{34b}$$

where $\nu_{\max}$ is the maximal viscosity. For both of these profiles a dimensionless vertical coordinate can be defined as

$$\tilde{z} = \frac{z}{H_{\text{BL}}} \implies \begin{cases} \nu_{\uparrow}(z) = \nu_{\max}\tilde{z} \\ \nu_{\downarrow}(z) = \nu_{\max}\left(1 - \tilde{z}\right). \end{cases} \tag{35}$$

Using these linear viscosity profiles, together with the new dimensionless vertical coordinate, Equation 8 is rewritten for the two profiles as

$$\frac{\mathrm{d}^2\tau}{\mathrm{d}\tilde{z}^2} = i \underbrace{\frac{1}{\tilde{z}}}_{\equiv 1/\tilde{\nu}_{\uparrow}} \underbrace{\frac{H_{\text{BL}}^2 f}{\nu_{\max}}}_{=2\varphi_{\text{BL}}^2} \tau, \tag{36a}$$

$$\frac{\mathrm{d}^2\tau}{\mathrm{d}\tilde{z}^2} = i \underbrace{\frac{1}{1-\tilde{z}}}_{\equiv 1/\tilde{\nu}_{\downarrow}} \underbrace{\frac{H_{\text{BL}}^2 f}{\nu_{\max}}}_{=2\varphi_{\text{BL}}^2} \tau. \tag{36b}$$

Above the dimensionless thickness φ was again introduced following the logic from before.

Another simple profile to evaluate is a parabolic increasing viscosity as

$$\nu_p = \frac{4\nu_{\max}}{H_{\mathrm{BL}}^2}\left(H_{\mathrm{BL}} - z\right) z. \tag{37}$$

Again, using the non-dimensional height coordinate $\tilde{z}$ and the dimensionless thickness φ, Equation 8 is rewritten as

$$\frac{\mathrm{d}^2\tau}{\mathrm{d}\tilde{z}^2} = i\,\underbrace{\frac{1}{4\tilde{z}\left[1-\tilde{z}\right]}}_{\equiv 1/\tilde{\nu}_p}\underbrace{\frac{H_{\mathrm{BL}}^2 f}{\nu_{\max}}}_{=2\varphi_{\mathrm{BL}}^2}(U - U_g). \tag{38}$$

Since the height coordinate is normalized for the height H_{BL}, the dimensionless thickness simply becomes the same as the inverse of the normalized Ekman scale height, defined as

$$\varphi = \frac{H_{\mathrm{BL}}}{h_{\mathrm{Ek(max)}}} = \frac{1}{\tilde{h}_{\mathrm{Ek(max)}}}. \tag{39}$$

3 Methodology

3.1 The numerical method

Equation 8 was solved numerically as a boundary value problem using the `solve_bvp` routine available in `SciPy` [Virtanen et al., 2020]. For the SUBC model, the equation was shown to be expressed in terms of a dimensionless layer thickness by using an height-normalized eddy viscosity profile, $\tilde{\nu}(z)$. Separating Equation 36a, 36b and 38 into zonal and meridional components yields

$$\frac{\mathrm{d}^2\tau_{\mathrm{x}}}{\mathrm{d}z^2} = -\frac{1}{\tilde{\nu}(z)}2\varphi_{\mathrm{BL}}^2\tau_y, \tag{40a}$$

$$\frac{\mathrm{d}^2\tau_{\mathrm{y}}}{\mathrm{d}z^2} = \frac{1}{\tilde{\nu}(z)}2\varphi_{\mathrm{BL}}^2\tau_x. \tag{40b}$$

To apply the numerical boundary value solver, the equations were rewritten as a first-order system by introducing the variables

$$\tau_x' = \frac{\mathrm{d}\tau_x}{\mathrm{d}z}, \qquad \tau_y' = \frac{\mathrm{d}\tau_y}{\mathrm{d}z}. \tag{41}$$

Expanding Equation 40 into four coupled first-order equations gives

$$\begin{cases} \frac{\mathrm{d}\tau_x}{\mathrm{d}z} &= \tau_x' \\ \frac{\mathrm{d}\tau_x'}{\mathrm{d}z} &= -\frac{2\varphi_{\mathrm{BL}}^2\tau_y}{\tilde{\nu}(z)} \\ \frac{\mathrm{d}\tau_y}{\mathrm{d}z} &= \tau_y' \\ \frac{\mathrm{d}\tau_y'}{\mathrm{d}z} &= \frac{2\varphi_{\mathrm{BL}}^2\tau_x}{\tilde{\nu}(z)}, \end{cases} \tag{42}$$

with the boundary conditions

$$\tau_x'(\varphi_{\mathrm{BL}} = 0) = 0, \qquad \tau_x(\varphi_{\mathrm{BL}} = 1) = 0, \tag{43a}$$

$$\tau_y'(\varphi_{\mathrm{BL}} = 0) = -fU_g, \qquad \tau_y(\varphi_{\mathrm{BL}} = 1) = 0. \tag{43b}$$

If the viscosity profile, $\tilde{\nu}$, includes a zero value in the domain, then the coupled Equation 42 becomes singular. We therefore add a constant background viscosity, ϵ

$$\hat{\nu} \to \hat{\nu} + \epsilon, \quad \text{where } \epsilon = \frac{\nu_{\text{background}}}{\nu_{\max}} \tag{44}$$

and $\hat{\nu}/\epsilon \gg 1$ is assumed. This background viscosity was thus applied for the linear decreasing, linear increasing, and the parabolic model since they have zero viscosity at the surface (or at the top).

The SUBC model equations were solved iteratively on a vertical grid with the imposed boundary conditions. The solver required an initial guess, which was chosen as the classical solution for a constant viscosity. The final solution was found when the iterative solution reached a steady state.

3.2 Analysis of wind-turning angles from ERA5

Reanalysed atmospheric data for different pressure levels was downloaded from the ERA5 dataset [Copernicus Climate Change Service, 2019] in order to observe the real-world surface wind angles. The data used an annual mean wind dataset for the year 2025. The analysis was restricted to the northern mid-latitudes in order to capture the Ekman spiral from the presence of the jet stream. In order to only analyse land points, a land mask was used from the package `Global_Land_Mask`, which is based on data from NOAA National Centers for Environmental Information [1999]. To avoid mountainous or very hilly terrain, an elevation dataset from NOAA National Centers for Environmental Information [2022] was used in order to dismiss points with a very steep linear slope. Angles with a standard deviation that was too large were also excluded. For the remaining grid points, the horizontal wind vectors were analysed. The surface wind vectors were compared to the wind at the location of the highest wind speed, i.e., the jet stream. The angle between these two vectors was computed as the signed angle between them by using the cross product. To avoid poorly defined angles, surface wind speeds below a threshold were discarded. The criteria for the data points can be summarised as follows:

- Northern mid-latitude band between $[30°, 60°]$.
- Land points only.
- Non-mountainous points with a linear slope below $1000\,\mathrm{m\,km^{-1}}$.
- Surface wind speeds above $1\,\mathrm{m\,s^{-1}}$.
- A standard deviation below 45°.

Lastly, a circular mean was calculated for the different times in order to obtain a average value for the temporal series. Thus, a geographical map together with angle distributions for the surface wind angles was obtained.

4 Result

4.1 Theoretical surface angles

4.1.1 The vertical structure of the momentum flux

The vertical profiles from the numerical SUBC model can be viewed in Figure 9–11 together with the normalized Ekman height for the different layers. An important note is that when $\varphi_{\mathrm{BL}} < 1$ then the Ekman heights become larger than the boundary layer height H_{BL}, which is an unphysical scenario. In contrast, if $\varphi_{\mathrm{BL}} \gg 1$, then the Ekman height is much smaller than the layer depth, $h_{\mathrm{Ek}} \ll H_{\mathrm{BL}}$, such that the spiral is closer ot the surface as $z \ll H_{\mathrm{BL}}$.

Since the surface wind angle is calculated by comparing the initial change of the viscosity, both the linear increasing and the parabolic models are dependent on the background viscosity ϵ. The effect of varying ϵ can be observed in Figure 10–11. However for the linear decreasing model, no major effect could be observed by varying the background viscosity. Thus, a background viscosity of 1×10^{-6} was used for this specific model.

4.1.2 The surface angles

Figure 12–14 show the surface angles from the numerical SUBC models. Similarities between these models, the decreasing two-layer model, and the exponentially decaying model can be observed. They all have a lower limit of 90° and an upper limit of 45°. The reason for the similarly between the the increasing

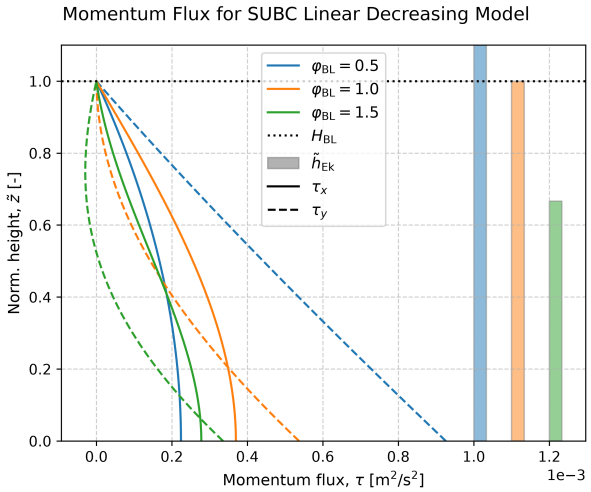


Figure 9: Numerically computed vertical structure for linearly decreasing viscosity. The momentum flux depends on the dimensionless parameter $\varphi_{\mathrm{BL}} = \tilde{h}_{\mathrm{Ek}}^{-1}$. The solution assumed a purely zonal geostrophic flow $U_g = 10\,\mathrm{m\,s^{-1}}$ and a Coriolis parameter of $f = 1 \times 10^{-4}\,\mathrm{s^{-1}}$.

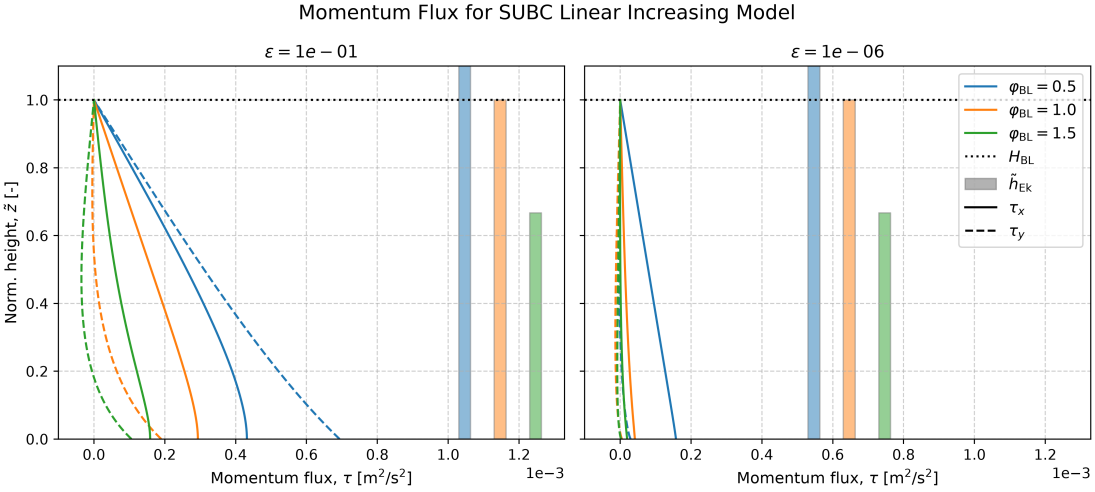


Figure 10: Numerically computed vertical structure for linearly increasing viscosity. The momentum flux depends on the dimensionless parameter $\varphi_{\mathrm{BL}} = \tilde{h}_{\mathrm{Ek}}^{-1}$ and the background viscosity ϵ. The solution assumed a purely zonal geostrophic flow $U_g = 10\,\mathrm{m\,s^{-1}}$ and a Coriolis parameter of $f = 1 \times 10^{-4}\,\mathrm{s^{-1}}$.

two layer model, is that the SUBC model is based on the decreasing two layer model, as $\nu = 0$ above the height H_{BL}. Again the linear behaviour of $\tau(z)$ when $\varphi_{\mathrm{BL}} \ll 1$ yields a surface angle of 90°, while the more constant viscosity when $\varphi_{\mathrm{BL}} \gg 1$ yields a surface angle of 45°. Thus, in the unphysical limit when $\varphi_{\mathrm{BL}} \to 0$ the Ekman height becomes infinitely larger then H_{BL} providing a 1.5 layer like solution which has the the limit of 90° and when $\varphi_{\mathrm{BL}} \to \infty$ the maximum value of the profile decreases, providing a smaller change with altitude, and a more constant like viscosity with a classical angle of 45°.

The result from the linearly decreasing SUBC model is shown in Figure 12, where the values smoothly transition from 90° → 45° as φ_{BL} increases. This plot resembles the exponential decaying model seen in Figure 6 and the 1.5 model as seen in Figure 2. Again, all angles are larger then the reference angle of 45°.

For the linearly increasing viscosity in Figure 13 a different result can be seen. The upper limit is again located at 45°, but at very large non-dimensional heights. For smaller background viscosities the limit occurs at many magnitudes higher then for the previous models due to a much larger $1/\tilde{\nu}_{\uparrow}$-factors.

The curve obtains values smaller then 45° outside of the limits discussed above. Since the lower limit is

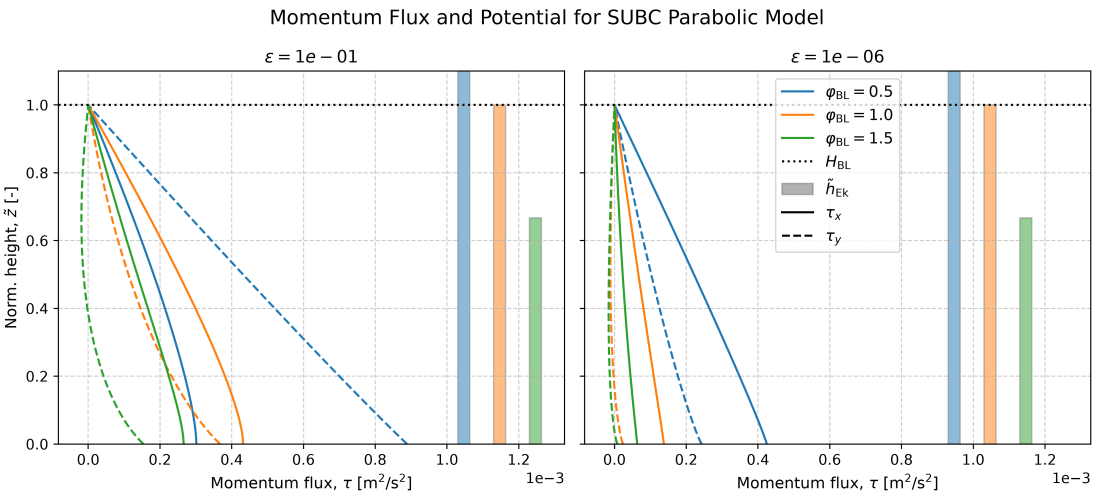


Figure 11: Numerically computed vertical structure for parabolic viscosity. The momentum flux depends on the dimensionless parameter $\varphi_{\mathrm{BL}} = \tilde{h}_{\mathrm{Ek}}^{-1}$ and the background viscosity ϵ. The solution assumed a purely zonal geostrophic flow $U_g = 10\,\mathrm{m\,s^{-1}}$ and a Coriolis parameter of $f = 1 \times 10^{-4}\,\mathrm{s^{-1}}$.

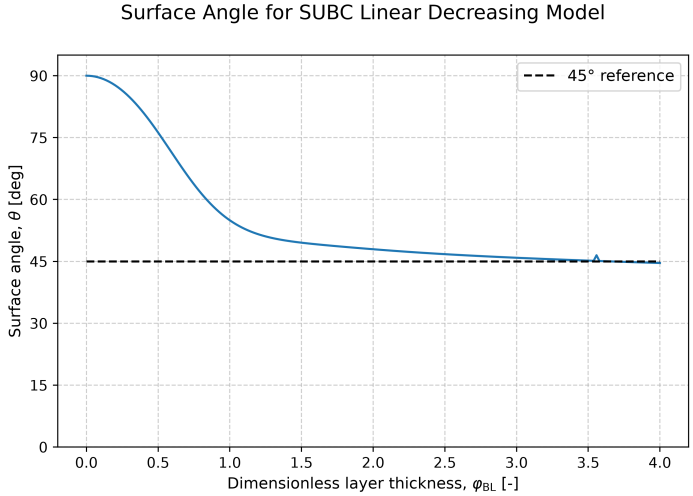


Figure 12: Numerically computed surface angles for decreasing viscosity profiles in the SUBC model. The angles are dependent on the dimensionless parameter $\varphi_{\mathrm{BL}} = \tilde{h}_{\mathrm{Ek}}^{-1}$.

an inherent property of the SUBC model and the upper limit is well understood, makes the interior more interesting. The surface wind angle is smaller then 45° since a larger viscosity at a higher altitude means that the rotation of the wind is smaller near the surface and greater aloft. This forces the surface angles to become less then 45°. This can be compared to the two-layer model with a small lower viscosity ν_1 and small interface height H_{BL}. Observing Figure 2, one can see that this scenario also yielded angles smaller than 45°, like in this figure.

Another interesting observation is that, away from the limits, she surface wind angle is more sensitive to the background viscosity than to the non-dimensional height. If the background viscosity is much smaller then the maximal viscosity then a smaller ϵ gives a huge $1/\tilde{\nu}_{\uparrow}$ jump from the surface. In the theoretical limit when $\epsilon \to 0$, the initial jump would be infinite, and the surface angle would be zero.

The SUBC model with a parabolic viscosity profile gave Figure 14. Very similar behaviour to the increasing viscosity profile in Figure 13 can be observed. Since the viscosity jump from the surface is what determines the surface angle, the parabolic decrees aloft has no dramatic effect on the surface angle. The two models differ slightly in that the parabolic curves cross the 45° later than the linearly increasing curves.

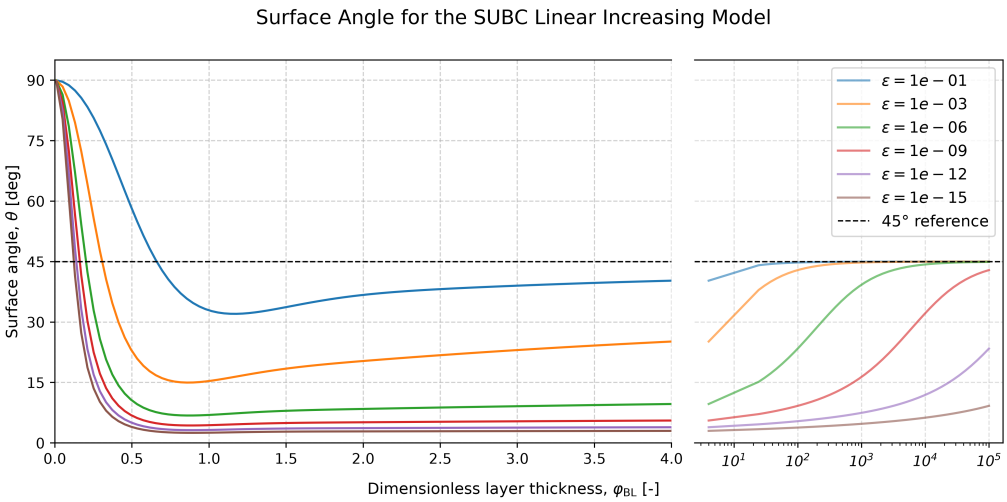


Figure 13: Numerically computed surface angles for linearly increasing. The angles are dependent on the dimensionless parameter $\varphi_{\mathrm{BL}} = \tilde{h}_{\mathrm{Ek}}^{-1}$ and the background viscosity ϵ.

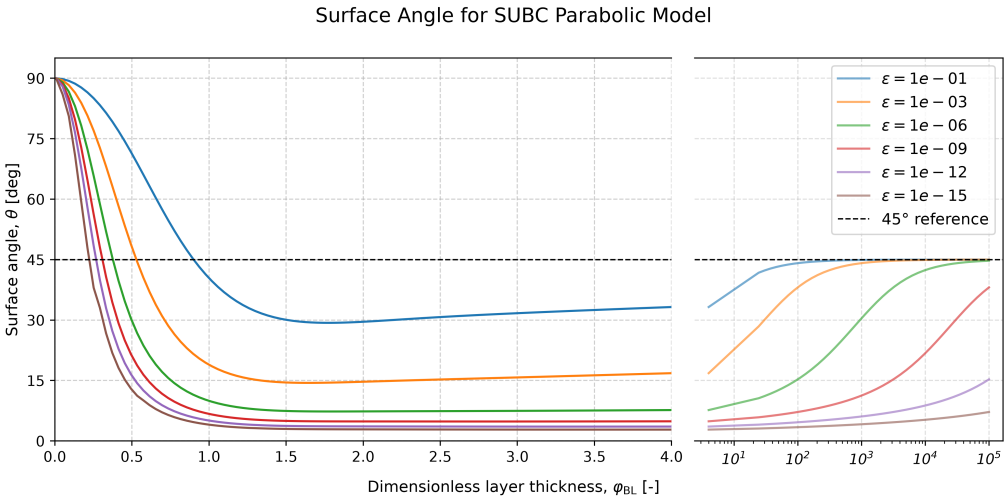


Figure 14: Numerically computed surface angles for a parabolic viscosity profile in the SUBC model. The values are dependent on the dimensionless parameter $\varphi_{\mathrm{BL}} = \tilde{h}_{\mathrm{Ek}}^{-1}$ and the background viscosity ϵ.

4.1.3 The Ekman transport

Computing the transport with the surface vertical momentum flux as Equation 12 yielded exactly the same shapes as seen in Figure 12–14, but centred around 135° rather then 45°, thus a difference of 90° like in the classical Ekman theory. The transport values in Figure 15–17 show similar behaviour with maximum zonal transport for the unphysical scenario when $\varphi_{\mathrm{BL}} \to 0$ and maximal meridional transport between $\varphi_{\mathrm{BL}} \in (0, 1)$.

4.2 ERA5 surface angles

14 346 data points were fit for analysis from the ERA5 dataset. These surface wind angles, due to the presence of a jet stream for non-mountainous land points in the northern midlatitudes, are shown in Figure 18. The geographical spread of the data points can be observed in Figure 18a, while Figure 18b shows the probability distribution. Figure 18b shows a wide range of surface angles, but with a clear peak between 0°–60°. Comparing this peak with the reference value of 45° shows that a clear majority of the surface angles are smaller than the reference, although a minority can be observed to be greater than the reference angle. The larger spread between -180° to 180° may arise from other mereological processes,

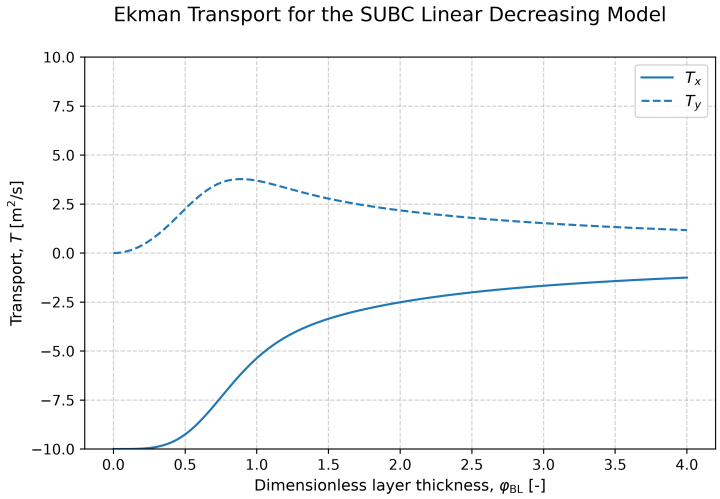


Figure 15: Numerically computed surface angles for a linear decreasing viscosity profile in the SUBC model. The values are dependent on the dimensionless parameter $\varphi_{\mathrm{BL}} = \tilde{h}_{\mathrm{Ek}}^{-1}$.

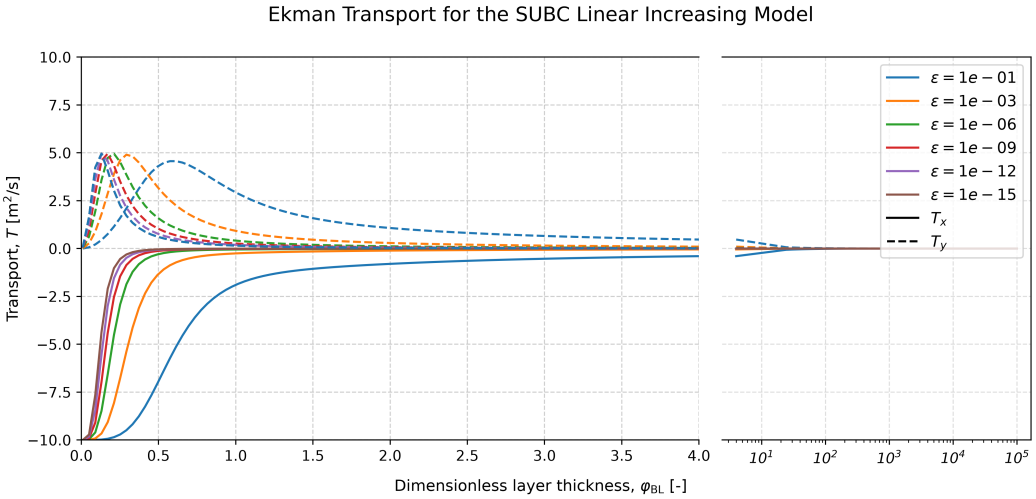


Figure 16: Numerically computed Ekman transport for a linear increasing viscosity profile in the SUBC model. The values are dependent on the dimensionless parameter $\varphi_{\mathrm{BL}} = \tilde{h}_{\mathrm{Ek}}^{-1}$ and the background viscosity ϵ.

such as the presence of a semi-permanent high- and low-pressure system, causing surface winds. Due to the smoothness of these values, these can be dismissed as outliers and the focus can lie on the peak between 0° to 60°.

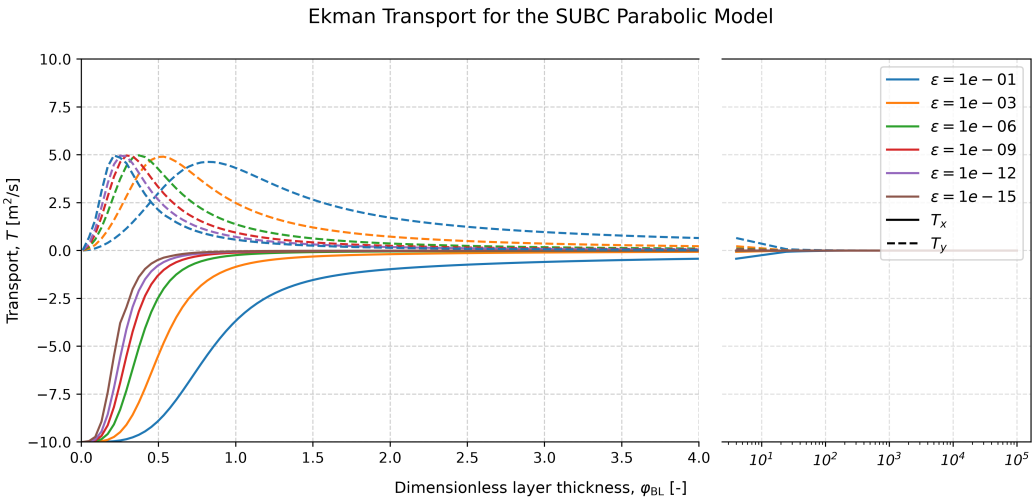
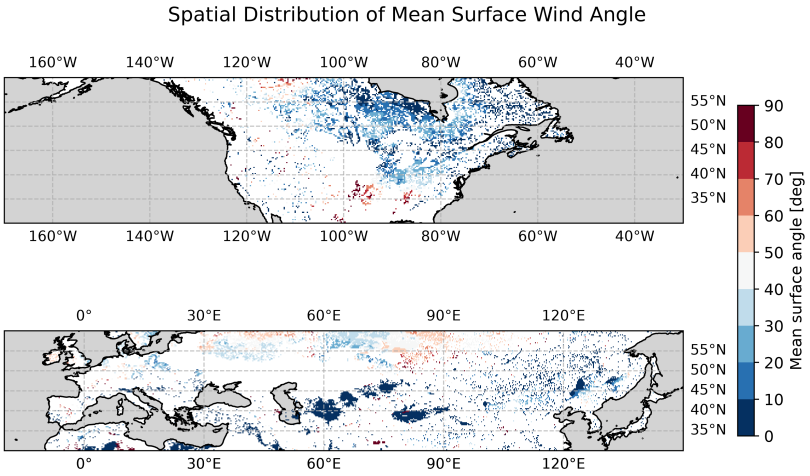
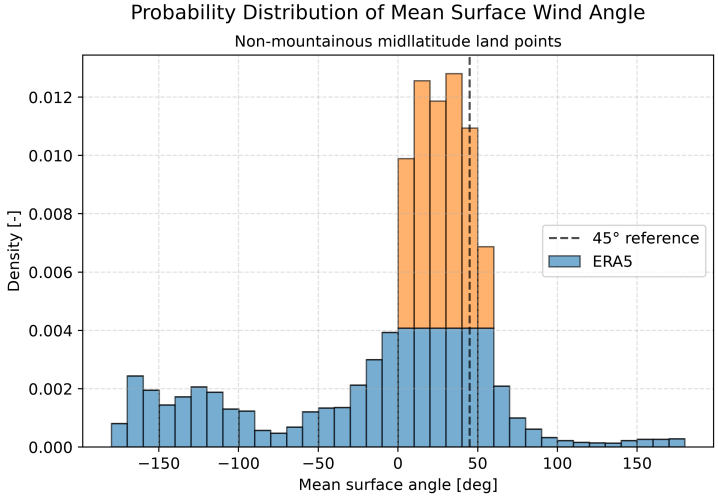


Figure 17: Numerically computed Ekman transport for a parabolic viscosity profile in the SUBC model. The angles are dependent on the dimensionless parameter $\varphi_{\mathrm{BL}} = \tilde{h}_{\mathrm{Ek}}^{-1}$ and the background viscosity ϵ.



(a) Mean surface wind angle over non-mountainous land points.



(b) Distribution of the mean surface wind angles. The peak is highlighted in orange.

Figure 18: Statistical analysis of the surface wind angles from ERA5 for non-mountainous land points located under a jet stream.

5 Discussion

From both analytical and numerical models, angles between 0° to 90° could be observed. Generally, surface wind angles above the classical value of 45° could be observed with strictly decreasing viscosities with altitude. The only case where this did not occur was for a small φ-region in the two-layer model, with a lowest value of 43.4°. The reasoning behind decreasing viscosities with altitude providing surface angles over 45° value can be attributed to decreasing viscosities forcing rotation to occur at a lower height. As the viscosity is initially large but decreases with height, the spiral is forced to turn more at the lower altitudes. This has to occur in order to satisfy the geostrophic limit above. Thus, the surface angle becomes larger than the classical value for the decreasing viscosities with altitude.

For the increasing viscosities with altitude, the opposite could be observed with smaller surface wind angles than the classical value. For the two-layer model, only a small region had angles greater than the classical value. However, the largest value was only measured at 46.6°, which is close to the classical limit. The same logic follows to explain why this surface angle became smaller, as the explanation for why the surface angles became larger for the decreasing viscosities with height. If the viscosity increases with height, then the spiral has to turn less near the surface in order to turn enough to reach the geostrophic limit aloft. And since the viscosity increases with altitude, more rotation can occur there, as it is the shear due to the viscosity which forces this rotation.

In the limit where the dimensionless layer thickness approaches infinity, the viscosity profile becomes constant, resulting in classical Ekman behaviour. Conversely, when the dimensionless layer thickness approaches zero, the extremes of the viscosity profiles emerge. This behaviour is primarily understood through the solution approaching linearity. For extreme viscosity ratios, the two-layer model explains this as the low interface height dominating the surface boundary conditions. Due to this linearity, a leading-order imaginary surface momentum flux arises when the upper Ekman height is small, while a leading-order real surface momentum flux occurs when the upper Ekman height is large relative to the interface.

Comparing the results from the theoretical models with the ERA5 data, a clear connection can be observed. The ERA5 data showed a clear majority of angles between 0° and 45°. As the atmosphere consists of a turbulent boundary layer viscosity generally increases with height, which connects the theoretical models to the observed data. However, a small portion of the data that had angles above 45°, although a boundary layer must be present. This can be attributed to the boundary layer having a small region where the viscosity decreases with height, or a strictly decreasing one producing slightly higher angels like in the two-layer model, which would allow surface angles above 45°. Although one should not forget that the ERA5 data are reanalysed data, and thus not pure observations, which also could explain this portion of the data. However, the bulk of the data is still in agreement with the theoretical models, supporting the findings in this article.

Comparing these results with previous works shows a general consistency. The general rule that strictly increasing viscosities with height produce surface angles below 45° and strictly decreasing viscosities with height produce surface angles above 45° can be compared to the previously mentioned earlier work. Ionescu-Kruse [2021] showed that by analysing two smoothly increasing polynomial viscosity profiles, only one of the two produced angles below 45°, with a quadratic profile producing angles above 45°. This was, however, corrected by Guan et al. [2023], who showed that the quadratic profile produced angles below 45°, agreeing with the findings in this article. The piecewise constant two-layer models by Dritschel et al. [2020] agreed with the findings here, that increasing viscosity yielded angles below 45° and vice versa. The perturbative studies by Bressan and Constantin [2019] and Constantin [2021] also support the findings in this article. Constantin [2021] highlighted that even strictly decreasing viscosities could yield angles below 45° and vice versa, like the two-layer model in this study. However, it seems like this can only occur for non-smoothly varying viscosities, as all the other models in this article and the previous works show that smoothly increasing viscosities yield angles below 45° and smoothly decreasing viscosities yield angles above 45°.

6 Conclusion

The two-layer model showed that strictly increasing viscosities with height mostly produced surface angles below 45°, while strictly decreasing viscosities mostly produced surface angles above 45°. The exponentially decaying model showed surface angles above 45°. The numerical model used a simplified upper boundary condition, which allowed for any viscosity profile to be studied. The linearly decreasing model gave surface angles above 45°, while the linearly increasing model and the parabolic model produced surface angles below, neglecting extremes caused by the new boundary condition. This showcased that the viscosity jump from the surface is what generally determines the surface angle, as increasing viscosities with height force the turning of the wind to occur aloft, while decreasing viscosities with height force the turning of the wind to occur near the surface. This was even true for the parabolic model, which had a decreasing viscosity aloft, giving angles below 45°. The analytical two-layer model could produce surface angles outside this rule, but only in a small region with lower magnitude angles.

From ERA5 data a clear majority of angles between 0° and 45° could be observed, supporting the atmosphere having an increasing viscosity near the surface, which agrees with the theory of the mixing length hypothesis and the boundary layer. This article has demonstrated how a combination of analytical, numerical, and reanalysed data can be used together to obtain a deeper understanding of the Ekman spiral and the effect of varying eddy viscosities with height. The results from this article can thus be used to further understand the Ekman spiral and thus get a deeper knowledge of the boundary layer in the atmosphere and ocean.

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A The Source Code

The source code for this article can be found on GitHub under:

`https://github.com/FredrikBergelv/Ekman-Spiral-Project`

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